

The Higgs Bridge Carrier: Kernel Consequences for Electroweak Mass-Response

Paper 1 of the Higgs Structural Endpoint Arc

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Abstract

This manuscript proves the first paper in the Higgs structural endpoint arc. Its target is not a numerical Higgs prediction, a derivation of the Standard Model Higgs sector, a new electroweak model, or a phenomenological fit. Its target is the local standing problem: what must be in force for a same-scope electroweak regime to support Higgs-domain claims as determinate, admissibility-bearing construction rather than as representative notation, empirical input, or downstream route success?

The answer is a kernel consequence, but the claim is conditional and target-relative. The paper first separates ordinary physics work from same-scope standing claims: a calculation, model, fit, diagnostic, or route may be valuable without claiming primitive Higgs standing. Once a fixed electroweak order-to-deformation and mass-response target is treated as claim-bearing, however, the four-role kernel

$$K_D = (K, \simeq_{\text{gov}}, I_D), \quad K = \{\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}\},$$

is already required. Reference fixes the electroweak target; standing fixes licensed continuation and reuse; irreversibility blocks same-act repair; and admissibility is the boundary under which these roles remain jointly enforceable. The imported admissibility-structure consequences are then specialized to the Higgs domain: the boundary is AMetric, the admissibility-relevant quotient is bivalent, standing is reuse-stable admissibility, same-scope generators/carriers/repairs are blocked, and fixed-domain exhaustion leaves no fifth local governance-bearing locus.

The manuscript defines a neutral Higgs-domain transition target D_H , a required role bundle H_{role} , and the canonical Higgs bridge/fixation carrier

$$C_H := H_{\text{br/fix}}.$$

It proves that scalarity, vev-status, potential-minimum status, pole-mass status, running-parameter status, Yukawa-form status, mass-matrix status, vacuum-branch status, and thermal-transition status are all representative or downstream route descriptions unless they descend through the Higgs bridge carrier and its standing-preserving quotient. The main positive theorem is the *Higgs bridge-carrier necessity theorem*: in a non-degenerate electroweak mass-response regime preserving gauge inheritance, vacuum-supported response reduction, deformation mediation, mass-response standing, identity continuity, and standing-preserving redescription, the neutral order-to-deformation transition forces a bridge-carrier normal form. The imported bridge-witness closure package then identifies the closed occupant as $C_H = H_{\text{br/fix}}$. The main negative theorem is the *primitive representative collapse theorem*: no familiar Higgs representative supplies primitive Higgs-domain standing by itself.

The endpoint closed here is the carrier theorem, not the full identity theorem of Paper 2. Paper 1 establishes the carrier normal form and the initial exhaustion register. The identity question is thereby isolated for Paper 2: what makes something *the Higgs* rather than merely Higgs-associated notation, scale, field, excitation, vacuum route, or extension content?

Keywords: AASC; Higgs bridge; kernel of admissibility; electroweak mass-response; Standard Model; gauge inheritance; vacuum-supported response; deformation mediation; representative collapse; Higgs identity; no-promotion; endpoint standing.

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Mathematical-status lock and proof-class boundary

This paper is an AASC fixed-target status theorem. It is not a Higgs-sector model, not a replacement for the Standard Model, not a derivation of a Higgs doublet, not a calculation of a vev or pole mass, not a new RG analysis, not a hierarchy solution, not a vacuum-stability calculation, and not an electroweak-baryogenesis model. Standard Higgs/electroweak language supplies the comparison arena. AASC supplies the status calculus under which target fixation, representative standing, carrier standing, no-promotion, same-scope exhaustion, and richer-scope update are separated.

The proof is evaluated inside the following proof class:

kernel import + AASC admissibility consequences + SM/Higgs bridge support corpus.

A same-target objection must invalidate one of the following: the kernel import, the admissibility consequence interface, the fixed Higgs-domain target, the required role bundle, the representative-insufficiency analysis, the carrier-normal-form construction, the bridge-carrier necessity theorem, the carrier-minimality theorem, the representative-collapse theorem, the status classifier, the exhaustion register, or the final survivor inference. A phenomenological recalculation or a different model-building preference is theorem-relevant only if it supplies, breaks, or updates one of those status-bearing objects.

Read this first: proof spine

The proof has six layers.

Layer I: kernel import. A claim-bearing same-scope Higgs-domain construction already requires the fixed-domain kernel

$$K_D = (K, \simeq_{\text{gov}}, I_D), \quad K = \{\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}\}.$$

A scalar symbol, vev coordinate, potential minimum, pole mass, running parameter, Yukawa placeholder, mass matrix, vacuum branch, or thermal profile cannot define the standing rule by which it wins.

Layer II: target lock. The paper fixes a neutral Higgs-domain target D_H whose job is to evaluate whether electroweak order and representation content can remain standing-bearing when used for deformation and mass-response claims. The target contains gauge inheritance, representation continuity, vacuum support, deformation-response slots, mass-response readiness, admissible redescription, and richer-scope update conditions. The target does not preselect a bridge carrier; it fixes the arena in which any mediator must be tested.

Layer III: role-bundle necessity. The required role bundle is

$$H_{\text{role}} = (\text{Gaugelnh}, \mathcal{N}_D^{\text{res}}, \text{VacSupp}, \text{DefMed}, \text{MassResp}, \text{IdCont}, \text{StandTrans}, \text{RedStab}).$$

Any same-scope Higgs-domain standing claim must preserve the bundle or declare target update. Omitting a component either loses the Higgs-domain target or imports hidden selector work.

Layer IV: representative insufficiency. Familiar Higgs representatives are admitted as useful descriptions, but none independently carries the whole role bundle. The proof classifies scalarity, vev-status, potential-minimum status, pole-mass status, RG/running status, Yukawa-form status, mass-matrix status, vacuum-branch status, and thermal-transition status as representatives or downstream route statuses.

Layer V: bridge-carrier normal form. Mediation alternatives are exhausted into representative-only, downstream-route, standing-smuggling, target-update, or bridge-carrier normal cases. The imported bridge-witness closure package identifies the closed bridge-carrier occupant as

$$C_H := H_{\text{br/fix}}.$$

Every same-scope primitive Higgs standing claim factors through the local Higgs normal form

$$R_H = (D_H, C_H, H_{\text{role}}, \mathcal{R}_H, \sim_H, \text{SF}_H).$$

Layer VI: initial endpoint register. The initial exhaustion register deletes primitive representative standing and preserves the bridge/fixation carrier as the only Paper-1 primitive survivor. The resulting carrier theorem supplies the formal input for the later Higgs structural identity endpoint.

Dependency-inversion exclusion

The dependency order is not

$$\text{AASC vocabulary} \Rightarrow \text{Higgs bridge conclusion}.$$

The dependency order is

$$\text{determinate Higgs-domain standing evaluation} \Rightarrow K_D \Rightarrow \text{Higgs bridge-carrier consequence layer}.$$

A Higgs representative cannot define the admissibility rule by which it becomes identity-bearing. Once a regime claims to evaluate scalar, vev, potential, pole-mass, running, vacuum, thermal, or extension descriptions as same-domain Higgs claims, target identity, standing, admissible redescription, and act-time finality are already in force. A contrary same-target construction must derive these roles from below without governance-equivalent work, or else concede the kernel under different vocabulary, or abandon the fixed Higgs-domain target.

1 Reader orientation: target scope and kernel-first discipline

This section fixes the interpretive contract of the paper before the formal Higgs argument begins. Its purpose is not to add a new premise. It makes explicit which target is being evaluated, what “degenerate” means inside that target, why the AMetric boundary is binary at the point of standing-bearing entry, and how ordinary physics work remains intact.

1.1 What “degenerate” means here

Definition 1.1 (Target-relative degeneracy). A regime, act, or claim is *degenerate relative to a target* when it purports to carry non-degenerate same-scope standing for that target while failing at least one of the following target-adequacy requirements: determinate target identity, admissible stephood, act-time finality, or identity-preserving redescription fidelity. Degeneracy is therefore a status relative to the claimed target. It is not a general criticism of a calculation, model, approximation, simulation, heuristic, or phenomenological fit.

Principle 1.2 (Conditional scope principle). The AASC constraints used in this paper govern claims of non-degenerate same-scope standing. They do not govern every useful piece of physics work. A calculation, ansatz, model, approximation, search trace, fit, diagnostic plot, or provisional construction may be valuable without claiming primitive same-scope standing. The constraints become active when such material is used as a standing-bearing Higgs-domain construction, identity ground, carrier selector, or endpoint-resolution act.

This principle is the main scope guardrail. The paper does not say that Standard Model calculations, effective potentials, RG analyses, vacuum-stability diagrams, or Higgs measurements are degenerate. It says that a representative or downstream route becomes subject to the kernel only when it is promoted into same-scope standing-bearing authority.

Table 1: Degeneracy is target-relative, not a criticism of ordinary use.

Object or practice	Ordinary status preserved here	When the AASC standing question begins
Perturbative calculation	legitimate calculation inside a declared scheme and error model	when its output is used as primitive same-scope standing rather than route or witness content
RG-improved potential	legitimate diagnostic and transport representative	when a displayed branch or scale is promoted into primitive Higgs standing
Collider measurement	empirical witness or boundary-normalized input	when the measured value is treated as identity ground or carrier selector
Model-building ansatz	legitimate exploration or richer-scope submission	when it is claimed as same-scope repair or primitive replacement without target update
Phenomenological fit	useful comparison and constraint	when fit success is treated as a generator of standing

1.2 Non-degenerate construction in one page

For this paper, non-degenerate construction is the package

target identity + admissible stephood + act-time finality + redescription fidelity.

The kernel terms name the governance roles forced by that package. Reference preserves target identity. Standing marks act-carried eligibility for licensed continuation and reuse. Irreversibility blocks later repair from counting as the same admissible act. Admissibility is the boundary under which those classifications remain jointly enforceable. The kernel source states this result as a fixed-domain exhaustion theorem for identity-preserving construction: raw traces and diagnostic grades may exist, but standing-bearing construction requires the kernel structure $K_D = (K, \simeq_{\text{gov}}, I_D)$ with $K = \{\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}\}$. [1]

A proof-assistant example gives the local shape. A failed tactic followed by a successful corrected tactic may produce a checked proof object. The corrected object can stand after checking. The failed tactic did not retroactively become the same admissible act. Treating it as such either changes act identity, imports repair authority, or reduces to bookkeeping about the history. The kernel source uses this trace/construction distinction to explain why later correction cannot generate same-act standing. [1]

A Higgs-domain example has the same form. A lower branch of an RG-improved effective-potential representative can be diagnostically meaningful, and a lifetime route can be available. But the representative branch does not thereby become a primitive endpoint. The vacuum-status paper separates lower-branch diagnostics, route availability, and standing closure by lifetime or richer-scope certificate. [12] Paper 1 uses the same discipline at the carrier level: scalar, vev, mass, RG, vacuum, and thermal descriptions may be admitted, but they do not generate the bridge/fixation carrier.

1.3 Why weaker governance fails

A same-scope alternative to the kernel has only five possible dispositions. It loses target identity, loses evaluable stephood, collapses act-time finality into repair, imports substitute governance, or performs the same governance work under another name. The kernel source and fixed-domain exhaustion source both treat this as an exhaustion result, not an informal list. [1, 3]

Table 2: Weaker-governance dispositions used throughout Paper 1.

Attempted weakening	Failure mode	Higgs-domain form
No fixed target	target loss	scalar, vev, mass, vacuum, and extension claims slide among different objects of evaluation
No standing-bearing step status	step-status loss	representatives are written but no act is licensed for same-domain continuation
Same-act repair allowed	repair collapse	later route success or precision is used to rescue the original primitive claim
Boundary selector added	illicit import	fit quality, naturalness, metric proximity, canonical representative, or route success becomes hidden authority
Alternative vocabulary only	governance-equivalent collapse	the same kernel work is performed while different words are used

This table explains the proof strategy used later. The paper does not ask whether a Higgs representative is useful. It asks whether the representative can preserve the Higgs-domain target while doing the governance work required for primitive standing. When it cannot, the representative remains available only in the role it can actually support: description, measurement, route, projection, target update, or bookkeeping.

1.4 The AMetric boundary: internal measures versus boundary authority

The AMetric claim is narrow. It does not deny that metrics, probabilities, likelihoods, error bars, naturalness measures, RG scales, or fit scores can be used internally after a target and gate are fixed. It denies that such structures can decide standing-bearing boundary entry while preserving the same fixed domain. The admissibility-structure paper proves this by separating the pre-admissible submission side from the evaluated side: no selector, grading, ranking, probability, typicality rule, canonical representative, or nontrivial metric has boundary authority in the pre-admissible role, while internal downstream structures can still be licensed after admission. [2, 5]

The boundary is binary because the admissibility-relevant quotient of evaluated status is binary: admitted or not admitted. Undefinedness remains possible as non-membership in the evaluated

Table 3: Internal use is allowed; boundary authority is not.

Structure	Internal downstream use	Boundary-authority use	Paper-1 classification
Fit quality or likelihood	allowed as empirical comparison	cannot generate carrier standing	witness or diagnostic only
Naturalness pressure	allowed as heuristic or sensitivity diagnosis	cannot select same-scope carrier	no primitive selector
RG scale or running value	allowed as transport coordinate	cannot license boundary entry	transport representative
Error bars or uncertainty envelope	allowed inside route certificate	cannot replace the gate	certificate component
Canonical representative	allowed after quotient choice	cannot choose admissibility before closure	skin unless witnessed
Metric/gravity comparison	allowed as later projection surface	cannot found Paper-1 carrier standing	downstream projection

fragment; it is not a third positive boundary value. Graded, probabilistic, or typicality language may survive as internal content only after the boundary has been fixed.

1.5 Standing cannot be earned downstream

The most common misreading is to treat successful calculation or empirical success as a generator of standing. Paper 1 uses the opposite discipline:

success, similarity, fit, route availability, or later precision $\nRightarrow$ primitive standing.

The impossibility suite compresses the consequence as no independent standing generators, no standing-transfer carriers, and no same-act repairs. [6] The phrase “no standing-transfer carriers” does not prohibit the Higgs bridge carrier C_H . It prohibits an evaluated act or route result from carrying standing across a boundary without fresh admissible closure.

In Higgs language, this means that a good scalar model, an accurate fit, a successful route calculation, a useful RG display, or a phenomenologically attractive extension may supply evidence, description, route content, or a new target. It does not by itself make the original representative into the primitive Higgs-domain carrier.

1.6 Relation to actual physics work

This manuscript is not a request that physics stop using its ordinary objects. The scalar field, vev, Higgs excitation, Yukawa terms, mass matrices, effective potentials, RG flows, vacuum-stability analyses, thermal profiles, and extension models remain part of the standard comparison language. The paper classifies what kind of standing those objects can have inside a fixed Higgs-domain target. Its contribution is therefore status-theoretic:

What role does this object play:

carrier, representative, route, witness, projection, update, or failed primitive?

That classification matters because later papers in the arc ask identity, scale, vacuum, extension, and synthesis questions. Paper 1 supplies the local carrier package those papers use. The broader arc

plan fixes the sequence: Paper 1 imports the full kernel, while Papers 2-6 work downstream of the Higgs carrier package. [13]

1.7 Reader-facing role map

The following role map is the shortest way to read the rest of the paper.

Table 4: Paper-1 role map for new AASC readers.

Role	Meaning in this paper	Common misreading blocked
D_H	neutral Higgs-domain transition target	the target already assumes the carrier
H_{role}	role bundle required for order-to-deformation and mass-response standing	the bundle is merely a list of desirable properties
C_H	bridge/fixation carrier closed by imported witness package	the carrier is an extra particle or a vev
$\mathcal{R}_H$	scalar, vev, mass, potential, RG, vacuum, thermal descriptions	representatives are being rejected as physics
R_H	local normal form through which same-scope standing claims factor	calculations are being denied unless written in AASC notation
$D_{H,+}$	declared richer-scope update	BSM models are forbidden

2 Introduction: the Higgs-domain standing problem

The Higgs domain is usually introduced through familiar objects: a scalar field, a potential, a vacuum expectation value, an excitation with a measured pole mass, Yukawa couplings, gauge-boson mass terms, running couplings, vacuum stability diagrams, finite-temperature transition profiles, and possible extensions with additional scalars or composite structure. Each object can be legitimate in its own calculational setting. The standing problem is that none of these objects, merely by being written or measured, determines what makes a Higgs-domain claim claim-bearing inside a fixed electroweak regime.

The first question is therefore not numerical. It is not: why is the weak scale what it is? It is not: what is the exact Higgs self-coupling? It is not: does the high-scale effective potential develop a lower region? Those questions are downstream. The prior question is:

what structural carrier must exist for Higgs-domain claims to have standing at all?

The answer developed here is carrier-level. A Higgs-domain primitive is not a scalar coordinate, not a vev, not a potential minimum, not a pole mass, not a running parameter, not a vacuum branch, and not a thermal transition. Those are representative or downstream route descriptions. The standing-bearing primitive closed in this paper is the Higgs bridge/fixation carrier C_H . It is the carrier by which gauge and representation order can be mediated into deformation and mass-response standing without losing target identity.

This paper is Paper 1 of a six-paper Higgs structural endpoint arc. Its role is to import the kernel and specialize the kernel consequences to electroweak mass-response. It does not prove the full identity

theorem. It builds the carrier normal form and the representative-collapse register that make the identity theorem possible.

2.1 Central distinction

The distinction closed by the manuscript is

$$\text{Higgs representative} \neq \text{Higgs carrier} \neq \text{Higgs identity endpoint}.$$

Representatives include scalar notation, vev coordinates, potential charts, pole masses, running parameters, Yukawa/mass-matrix forms, vacuum diagnostics, and thermal profiles. The carrier is $C_H = H_{\text{br/fix}}$. The identity endpoint is reserved for Paper 2, where the question is whether Higgs identity is exhausted by unique occupation of that carrier role.

2.2 Main output

The main theorem package can be compressed as follows:

$$\begin{aligned} \text{ClaimBearingUse}_{D_H}(C) &\Rightarrow K_D, \\ \text{SameScopeHiggsStanding}(C) &\Rightarrow C \text{ factors through } R_H, \\ \text{Surv}_{D_H}^{\text{Primitive}}(C_{\text{scalar}}) &= \text{Surv}_{D_H}^{\text{Primitive}}(C_v) = \text{Surv}_{D_H}^{\text{Primitive}}(C_V) = \text{Surv}_{D_H}^{\text{Primitive}}(C_{m_H}) = 0, \\ \text{Surv}_{D_H}^{\text{Primitive}}(C_H) &= 1 \quad \text{inside Paper 1's carrier target.} \end{aligned}$$

The final line is not a full uniqueness-of-Higgs-identity theorem. It means that in the Paper-1 target class, the only primitive survivor is the bridge/fixation carrier role. Paper 2 must then prove that no same-scope alternative occupant can preserve the full identity bundle without reducing to C_H .

2.3 What the paper does not prove

Nonclaim 2.1 (No numerical Higgs prediction). This paper does not derive the Higgs vev, Higgs pole mass, Higgs self-coupling, total width, decay rates, weak mixing angle, Yukawa matrices, vacuum lifetime, or hierarchy magnitude.

Nonclaim 2.2 (No derivation of the Standard Model). This paper does not derive the Standard Model carrier inventory. It uses the Standard Model structural support as a comparison arena and extracts a Higgs-domain status theorem.

Nonclaim 2.3 (No anti-extension theorem). This paper does not rule out singlet scalars, two-Higgs-doublet models, supersymmetry, composite Higgs models, SMEFT descriptions, dilaton-like sectors, or richer UV completions. It classifies them as representative additions, same-carrier realizations, or target updates depending on their carrier and descent certificates.

3 Kernel import and consequence machinery

3.1 Kernel import

Definition 3.1 (Fixed-domain kernel structure). For a fixed domain D , the imported kernel structure is

$$K_D = (K, \simeq_{\text{gov}}, I_D), \quad K = \{\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}\}.$$

Here **Adm** is the boundary of joint enforceability, **Stand** is act-carried eligibility for licensed continuation and reuse, **Ref** preserves target identity, **Irr** blocks same-act repair, $\simeq_{\text{gov}}$ is governance equivalence on the fixed domain, and I_D is the admissibility-relevant invariant bundle.

Theorem 3.2 (Kernel import theorem). *The kernel paper proves that any target-adequate, non-degenerate fixed-domain construction regime determines the kernel structure K_D up to governance equivalence. AASC names the admissibility/standing-centered interface of that kernel and does not replace reference or irreversibility. [1]*

Proof. This is an imported theorem. The present paper does not reprove the kernel exhaustion theorem. It uses the kernel as the primitive governance floor for Higgs-domain standing evaluation. $\square$

Corollary 3.3 (Higgs-domain kernel trigger). *If a same-scope electroweak regime treats a Higgs-domain claim C as claim-bearing rather than as mere notation, then*

$$\text{ClaimBearingUse}_{D_H}(C) \Rightarrow K_D.$$

Proof. A claim-bearing Higgs-domain act must fix what target is being evaluated, which descriptions are licensed, which transitions preserve the same claim, and why later correction or extension cannot retroactively make the same failed act standing-bearing. These are exactly reference, admissibility, standing, and irreversibility. The kernel import theorem applies. $\square$

3.2 Admissibility consequence interface

The structure-of-admissibility manuscript supplies the dedicated proof of the admissibility-side normal form once the kernel is imported: AMetric submission boundary, bivalent admissibility-relevant quotient, standing as reuse-stable admissibility, uniqueness of the admissible interior, and conservation of standing. [2]

Theorem 3.4 (Admissibility consequence import). *Inside the fixed-domain kernel target class, the following hold for admissibility-relevant Higgs-domain status:*

- (a) *the pre-admissible boundary is AMetric;*
- (b) *the admissibility-relevant quotient of evaluated status is bivalent;*
- (c) *standing is reuse-stable admissibility;*
- (d) *the admissible interior is unique up to identity-preserving transformation;*
- (e) *standing is conserved: it is preserved or destroyed, not created, amplified, transferred, or repaired internally.*

Proof. Imported from the admissibility-structure theorem. The present paper specializes these consequences to Higgs-domain representative claims. $\square$

3.3 Fixed-domain exhaustion and operator closure imports

The fixed-domain exhaustion theorem states that same-scope admissibility-bearing variation must alter boundary typing, the standing partition, the lawful quotient, or admissible continuation; there is no fifth same-scope governance-bearing locus. [3] Closure by exhaustion for same-scope operators proves that same-scope operators on a standing-bearing carrier are bookkeeping-equivalent or inadmissible. [4]

Theorem 3.5 (No fifth local Higgs governance locus). *For the Higgs-domain target D_H , any same-scope admissibility-bearing variation must alter at least one of:*

- (i) *boundary typing or evaluated-fragment membership;*
- (ii) *the Higgs-domain standing/admissibility partition;*
- (iii) *the lawful Higgs representative quotient;*
- (iv) *admissible continuation or route structure.*

No fifth same-scope governance-bearing locus remains.

Proof. This is the Higgs specialization of fixed-domain exhaustion. If a submitted distinction changes none of the four loci, it changes no admissibility-bearing fact and is bookkeeping. If it changes something else, it has changed the comparison class or imported a new boundary primitive and is a target update or fail-closed attempt. $\square$

3.4 No independent standing generator, no transfer-carrier, no repair import

The impossibility suite records the compressed consequences of binary gate-produced standing and conservation: no independent standing generators, no standing-transfer carriers, and no same-act repairs. [6] This does not forbid standing-bearing carriers such as C_H . It forbids an evaluated act, representative, empirical success, or route-success report from carrying standing across the boundary without admissible closure.

Principle 3.6 (No Higgs-domain standing laundering). Technical success, explanatory adequacy, empirical fit, naturalness pressure, calculational convenience, later precision, or downstream route availability cannot generate, transfer, or repair Higgs-domain standing. Any such use is either bookkeeping, a fresh gated act, or target update.

4 The fixed Higgs-domain target D_H

Definition 4.1 (Neutral Higgs-domain transition target). The fixed Higgs-domain target D_H is the same-scope electroweak order-to-deformation comparison target containing:

1. a gauge/representation structure whose response must be preserved under admissible redescription;
2. a residual/vacuum support ledger sufficient for vacuum-mediated response reduction;
3. deformation-response standing slots, including mass-response readiness;
4. a neutral transition problem: order/representation standing must remain evaluable when used for deformation and mass-response claims, without preselecting the mediator that performs that transition;
5. representative descriptions: scalar fields, vev-like coordinates, potential charts, pole-mass records, running parameters, Yukawa-form slots, mass matrices, vacuum diagnostics, and thermal profiles;
6. a same-scope/richer-scope boundary for additional scalar, threshold, composite, UV, gravitational, cosmological, or source-state structure.

The target contains the comparison arena, the transition burden, and the candidate descriptions. It does not define the Higgs carrier by fiat.

Definition 4.2 (Order-to-deformation transition). The order-to-deformation transition is the admissibility-relevant passage from gauge/representation order to deformation-response claims, including mass-response claims, while preserving identity, residual normal form, and same-scope redescription stability.

Definition 4.3 (Mass-response readiness). Mass-response readiness is the status of a carrier or route that licenses mass-response slots as admissible deformation-response positions without supplying downstream numerical masses, empirical fits, or phenomenological calibrations.

Remark 4.4 (Relation to Standard Model support). The Standard Model structural source supplies the fixed carrier architecture in which gauge carriers, representation content, vacuum/bridge roles, Yukawa readiness, mass-response slots, and Higgs-like bridge/fixation content are treated as structural positions rather than as numerical predictions. [10]

Definition 4.5 (Same-scope Higgs-domain claim). A claim C is a same-scope Higgs-domain claim in D_H if it preserves the target inventory, gauge/representation ledger, residual/vacuum support, deformation-response target, admissible redescription rules, and boundary between licensed representative use and target update.

Definition 4.6 (Richer-scope Higgs update). A richer-scope Higgs update $D_H \rightarrow D_{H,+}$ occurs when a claim adds, removes, or changes carrier, threshold, scalar inventory, compositeness structure, symmetry, UV boundary, gravitational rule, cosmological source, or empirical witness in a way that changes the admissibility-relevant invariant bundle.

Proposition 4.7 (Context non-smuggling). *The definition of D_H does not assert C_H as the selected carrier. It fixes the arena, transition burden, and representative catalogue in which any submitted mediator must preserve the required role bundle under the imported kernel constraints.*

Proof. The target definition lists required comparison ledgers, the neutral order-to-deformation transition burden, and candidate descriptions. It does not delete representatives as useful descriptions, and it does not infer carrier standing. Carrier standing is proved only after role-bundle necessity, mediation-alternative exhaustion, representative insufficiency, imported bridge-witness closure, and normal-form factorization. $\square$

5 The required Higgs role bundle

Definition 5.1 (Higgs role bundle). The required Higgs-domain role bundle is

$$H_{\text{role}} = (\text{Gaugelnh}, \mathcal{N}_D^{\text{res}}, \text{VacSupp}, \text{DefMed}, \text{MassResp}, \text{IdCont}, \text{StandTrans}, \text{RedStab}),$$

where:

Gaugelnh gauge and representation inheritance are preserved;

$\mathcal{N}_D^{\text{res}}$ residual normal form is not reopened by a scalar representative;

VacSupp vacuum-supported response reduction is available and gauge/vacuum compatible;

DefMed the order-to-deformation transition is mediated without changing target or promoting representatives;

MassResp mass-response readiness is licensed without importing downstream mass values;

IdCont identity continuity is preserved under same-domain redescription;

StandTrans standing-bearing eligibility is preserved across the order-to-deformation transition without representative promotion or scope drift;

RedStab admissible redescription preserves the claim's target, role, and quotient status.

Lemma 5.2 (Role-bundle necessity). *Any same-scope Higgs-domain standing claim must preserve every component of H_{role} , or else declare target update.*

Proof. If gauge inheritance fails, the claim no longer evaluates the same electroweak/gauge target. If residual normal form fails, the claim reopens an upstream residual/vacuum ledger rather than operating within the fixed target. If vacuum support fails, the putative Higgs role lacks the response-reduction context in which the order-to-deformation transition can be meaningful. If deformation mediation fails, the claim does not connect order/representation standing to deformation or mass-response standing. If mass-response readiness fails, the transition is not relevant to the Higgs-domain role under evaluation. If identity continuity fails, same-scope use has been replaced by presentation drift. If standing-preservation across the transition fails, the claim attempts to move standing by pipeline, fit, or notation instead of preserving standing-bearing eligibility across the transition. If redescription stability fails, the claim is representative skin. Thus each component is forced by the target; omitting a component either loses the target or changes scope. $\square$

Corollary 5.3 (No one-property Higgs role). *No single familiar property - scalarity, vev-status, potential minimization, pole-mass status, RG running, Yukawa coupling, mass-matrix participation, vacuum-branch behavior, or thermal-transition behavior - can by itself supply H_{role} .*

Proof. Each named property can occur while at least one component of H_{role} remains unspecified: for example scalarity does not fix mass-response standing; a vev-like coordinate does not fix gauge-invariant bridge standing; a pole mass is a response witness rather than a carrier; and a vacuum branch is a route diagnostic rather than bridge/fixation identity. Therefore no one property supplies the full role bundle. $\square$

6 Representative insufficiency

6.1 Representatives admitted

Definition 6.1 (Higgs representative). A Higgs representative is a presentation, coordinate, calculation object, diagnostic, response record, or downstream route object used to describe Higgs-domain structure inside D_H . The representative class includes scalar field symbols, potential coordinates, vev-like coordinates, pole-mass records, RG/running labels, Yukawa expressions, mass matrices, vacuum branches, and thermal profiles.

Principle 6.2 (Positive admission of representatives). The paper admits Higgs representatives as legitimate descriptions when they are used through the carrier/quotient discipline of D_H . It denies only primitive standing for a representative as such.

6.2 Insufficiency table

Table 5: Representative insufficiency register for Paper 1.

Representative	Legitimate role	Why it is insufficient as primitive standing
Scalar field symbol	Field-theoretic presentation of a scalar sector or bridge candidate	Scalarity does not by itself preserve gauge inheritance, vacuum support, deformation mediation, or mass-response standing.
Vev-like coordinate	Boundary-normalized or chart-level description of response baseline	A vev coordinate can be gauge, scheme, normalization, or input-set dependent unless extracted through a standing carrier. It cannot select the carrier it later represents.
Potential minimum	Chart or effective-potential presentation of scalar response	Potential charts and off-shell minima can vary with gauge, scheme, field coordinate, and normalization; endpoint standing requires quotient descent.
Pole mass	Measured excitation or response record	A pole mass is witness-bearing response content, not the primitive bridge by which gauge/representation order becomes deformation response.
Running parameter or RG scale	Transport coordinate and diagnostic	RG values transport standing-bearing inputs; they do not select the Higgs carrier or supply primitive identity.
Yukawa form	Downstream compatibility slot for fermion mass-response	Yukawa compatibility presupposes a bridge carrier and cannot select the bridge at Paper 1.
Mass matrix	Downstream deformation-response object	Mass matrices require bridge, Yukawa, rank, and diagonalization ledgers; they are not primitive bridge/fixation standing.
Vacuum branch	Static or lifetime diagnostic	Vacuum branch standing belongs to the later vacuum endpoint route; it does not define the bridge carrier.
Thermal transition	Finite-temperature route status	Thermal order is a downstream thermodynamic target, not the role by which the Higgs carrier is fixed.

Table 6: Row-level representative failure certificate.

Representative	Failed primitive-role test	Failure mode if promoted as primitive	Admitted status
Scalarity	DefMed, MassResp, RedStab	scalar notation can be written without preserving gauge-to-deformation standing or quotient-stable redescription	representative
Vev-like coordinate	StandTrans, RedStab	coordinate, gauge, normalization, or input-set dependence becomes a selector if used before closure	boundary representative
Potential minimum	RedStab, StandTrans	field-coordinate or effective-potential chart is promoted into endpoint authority without quotient descent	diagnostic or route input
Pole mass	DefMed, StandTrans	measured response witness is made to generate the bridge it presupposes	response witness
Running parameter/RG scale	RedStab, StandTrans	scale or scheme coordinate is treated as boundary authority rather than transport diagnostic	transport diagnostic
Yukawa form	DefMed, MassResp	downstream compatibility is used to select the upstream transition role	compatibility slot
Mass matrix	DefMed, MassResp	diagonalization/rank data are used before the deformation-response ledger is licensed	response ledger
Vacuum branch	VacSupp, RedStab	branch diagnostic or lifetime route is used as carrier identity	vacuum route
Thermal transition	VacSupp, RedStab	finite-temperature route status is used as same-scope carrier standing	thermal route

Theorem 6.3 (Representative insufficiency theorem). *No representative in Table 5 supplies primitive Higgs-domain standing in D_H . Each representative becomes standing-bearing only by descending through C_H , a quotient-stable route, or a declared target update.*

Proof. Let R be one of the listed representatives. Table 6 records a failed primitive-role test for each row. If R is used only as a presentation of a carrier already preserving H_{role} , then R is not primitive; its standing is inherited through the carrier. If R is used to perform route-specific work - vacuum

lifetime, thermal transition, hierarchy display, mass-matrix classification, or extension descent - then R belongs to a downstream route and is not the bridge primitive. If R is used to select Higgs standing before the carrier or route is fixed, it functions as an independent standing generator, transfer-carrier, or boundary selector. The AMetric boundary, bivalence, no independent standing generator/no transfer-carrier/no same-act repair principles, and fixed-domain exhaustion block that move. Thus each representative is admitted only as carrier description, route content, bookkeeping, or target update, never as primitive same-scope Higgs-domain standing. $\square$

Corollary 6.4 (No primitive vev selector). *A vev-like coordinate cannot select Higgs-domain standing. It may only represent an already admitted bridge/fixation carrier or serve as boundary-normalized empirical input in a downstream comparison.*

Proof. A vev-like coordinate by itself does not fix H_{role} . If it is used to select the carrier, it reverses dependency order. If it is extracted from a carrier, it is no longer primitive. If it is measured or reconstructed, it is boundary-normalized input. The primitive selector route is therefore closed. $\square$

Corollary 6.5 (No primitive scalar-field identity). *A scalar field symbol or scalar multiplet presentation is not Higgs-domain identity. Scalar notation may present C_H after carrier standing is fixed, but cannot define C_H .*

7 The Higgs bridge/fixation carrier

Definition 7.1 (Higgs bridge/fixation carrier). The Higgs bridge/fixation carrier is

$$C_H := H_{\text{br/fix}},$$

the standing-bearing carrier mediating gauge/representation order into deformation and mass-response standing over the fixed Higgs-domain target. It is not a vev, not a pole mass, not a potential coordinate, not a running parameter, not a Yukawa matrix, not a vacuum branch, and not a thermal verdict.

Remark 7.2 (Relation to MGN3 bridge support). The Higgs-like bridge paper establishes that bridge standing attaches to a canonical bridge carrier obtained from bridge witnesses, response signature, reduced mediation map, and bridge role-rank, while vev-like coordinates, potentials, normalizations, subgroup labels, empirical scales, Yukawa placeholders, mass matrices, and mixing data cannot select the bridge. [8]

Definition 7.3 (Bridge witness package). A Higgs bridge witness package in D_H is a tuple

$$W_H = (W_D^{\text{br}}, Q_D^{\text{ord}}, Q_D^{\text{def}}, \mathcal{N}_D^{\text{res}}, \text{GVac}_D, \text{DefMed}, \text{MassResp}),$$

where W_D^{br} records bridge witnesses, Q_D^{ord} and Q_D^{def} are order and deformation quotients, $\mathcal{N}_D^{\text{res}}$ fixes residual normal form, GVac_D records gauge-vacuum compatibility, DefMed records mediation, and MassResp records mass-response readiness.

Theorem 7.4 (Imported bridge-witness closure certificate). *The MGN3 and Standard Model structural support package identifies the closed Higgs bridge occupant as the quotient-reduced bridge witness class, response signature, reduced mediation map, and bridge role-rank that preserve residual normal form, gauge-vacuum compatibility, deformation mediation, and mass-response readiness. In the notation of this paper, that closed occupant is denoted*

$$H_{\text{br/fix}}.$$

Table 7: Imported closure dependencies for the bridge-witness certificate.

Imported component	Support source	Work performed in Paper 1
Selector exclusion	MGN3 bridge support	Excludes vev-like coordinates, potentials, normalizations, subgroup labels, empirical scales, Yukawa forms, mass matrices, and mixing data as primitive bridge selectors.
Bridge witness package	MGN3 bridge support	Supplies the quotient-reduced bridge witness class, response signature, reduced mediation map, and bridge role-rank after selector deletion.
Standard Model structural instantiation	Standard Model structural support	Locates the bridge/fixation role inside the electroweak structural tuple without treating phenomenological success as primitive authority.
Residual normal-form preservation	Kernel, fixed-domain, closure, and residual-support sources	Prevents scalar, potential, vacuum, or operator notation from reopening the reduced target after admissible closure.
Gauge-vacuum compatibility	Gauge-vacuum support	Ensures that bridge/fixation is gauge-compatible response mediation rather than isolated scalar notation.
Mass-response readiness	Standard Model and bridge support	Licenses mass-response slots as admissible deformation-response positions without importing downstream mass values.

Proof. This is an imported bridge-support theorem, not a new phenomenological derivation. The dependencies in Table 7 specify the imported work. Paper 1 does not reprove selector exclusion, bridge-witness construction, Standard Model tuple instantiation, residual normal-form preservation, gauge-vacuum compatibility, or mass-response readiness. It uses those facts only after the neutral target, role-bundle necessity, representative insufficiency, and mediation-exhaustion analysis force a bridge-carrier normal form. The imported package supplies occupant closure for that form; it does not supply primitive empirical authority. $\square$

Criterion 7.5 (Carrier admission). A candidate B is admitted as the same-scope Higgs bridge/fixation carrier only if:

1. it preserves D_H and I_D ;
2. it supplies or factors through a bridge witness package W_H ;
3. it preserves every component of H_{role} ;
4. it survives admissible redescription under $\sim_H$;
5. it imports no representative selector and no downstream mass/vacuum/thermal success as carrier authority.

Theorem 7.6 (Carrier admissibility theorem). *Under the imported bridge-witness closure certificate, the carrier $C_H = H_{\text{br/fix}}$ is an admissible same-scope Higgs-domain carrier in D_H .*

Proof. The neutral target and role-bundle analysis determine what a same-scope mediator must preserve. The representative insufficiency theorem excludes scalar, vev, potential, pole-mass, RG, Yukawa, mass-matrix, vacuum, and thermal representatives as primitive selectors. The imported bridge-witness closure certificate supplies the closed occupant of the remaining bridge role, namely

$H_{\text{br/fix}}$. Since the carrier criterion requires preservation of D_H , I_D , H_{role} , and $\sim_H$, and since the imported certificate supplies the bridge witness package without representative selector authority, $C_H = H_{\text{br/fix}}$ is admitted as the Higgs bridge/fixation carrier. $\square$

8 Bridge necessity

Definition 8.1 (Non-degenerate electroweak mass-response regime). A Higgs-domain target D_H is non-degenerate for electroweak mass-response if it contains a nontrivial gauge/representation structure, a vacuum-supported response-reduction ledger, a deformation-response target, and at least one standing-bearing mass-response slot whose admissibility depends on preserving identity under same-domain redescription.

Lemma 8.2 (Bridge locus). *In a non-degenerate electroweak mass-response regime, the standing-bearing transition from gauge/representation order to deformation/mass-response content cannot be supplied by gauge notation alone, scalar notation alone, empirical scale input alone, or downstream mass fit alone.*

Proof. Gauge notation alone preserves representation language but does not mediate deformation or mass-response standing. Scalar notation alone can be written without preserving gauge inheritance or residual normal form. Empirical scale input supplies boundary-normalized values but not same-domain standing. Downstream mass fit presupposes the transition by which mass-response slots are licensed. Therefore the transition locus requires a non-representative mediator if it is to remain same-scope and standing-bearing. $\square$

Lemma 8.3 (Mediation alternative exhaustion). *Every same-scope protocol that purports to mediate gauge/representation order into deformation or mass-response standing in D_H falls into one of five status classes:*

RepresentativeOnly, RouteNormal, StandingSmuggling, TargetUpdate, CarrierNormal.

No sixth same-scope mediation class remains.

Proof. Let P be a submitted mediator. If P supplies only scalar notation, a vev coordinate, a potential chart, a pole-mass record, an RG label, a Yukawa expression, a mass matrix, a vacuum branch, or a thermal profile, then P is **RepresentativeOnly**. If P 's force depends on a certified hierarchy, vacuum, thermal, mass-matrix, extension, or later route map over an already admitted carrier, then P is **RouteNormal** rather than primitive mediation. If P treats representative availability, empirical success, calculational convenience, naturalness pressure, or later precision as standing before closure, then P is **StandingSmuggling**. If P changes carrier inventory, source state, threshold structure, UV boundary, symmetry, compositeness structure, gravitational rule, cosmological condition, or the invariant bundle I_D , then P is **TargetUpdate**. If none of these failure or downstream statuses applies and P preserves D_H , H_{role} , I_D , and redescription stability, then P occupies the bridge-carrier normal form. Fixed-domain exhaustion excludes any remaining same-scope governance-bearing locus outside boundary typing, standing partition, lawful quotient, and admissible continuation, so no sixth mediation class remains. $\square$

Thus the no-sixth-class claim is not a phenomenological inventory claim. It is the fixed-domain exhaustion theorem applied to the possible standing locations for a same-scope mediator.

Table 8: Mediation classes as fixed-domain loci.

Status class	Fixed-domain locus	Reason no further class remains
RepresentativeOnly	Standing partition	Descriptive availability is separated from standing authority.
RouteNormal	Admissible continuation	Downstream route maps presuppose admitted carrier standing.
StandingSmuggling	Boundary typing and standing partition	Selector authority is imported before admissibility closure.
TargetUpdate	Boundary typing	The submission changes the domain rather than preserving same scope.
CarrierNormal	Lawful quotient and admissible continuation	The submission preserves the role bundle and survives redescription.

Theorem 8.4 (Higgs bridge-carrier necessity). *Let D_H be a non-degenerate same-scope electroweak mass-response regime preserving the kernel, the admissibility consequence interface, and the required role bundle H_{role} . Then any same-scope primitive Higgs-domain standing claim must factor through a bridge-carrier normal form. Under the imported bridge-witness closure certificate, the closed occupant of that form is $C_H = H_{\text{br/fix}}$.*

Proof. By non-degeneracy, the regime contains standing-bearing gauge/representation content and standing-bearing deformation/mass-response content. By H_{role} , a Higgs-domain standing claim must preserve the order-to-deformation transition while retaining residual normal form, vacuum support, identity continuity, standing-bearing eligibility, and redescription stability. The bridge locus lemma excludes bare gauge notation, scalar notation, empirical scale input, and downstream success as sufficient mediators. Lemma 8.3 then exhausts all submitted mediators: representative-only protocols lack primitive standing; downstream routes presuppose a carrier rather than replacing it; standing-smuggling is blocked by the kernel consequences; and target updates leave the same-scope target. The only surviving same-scope primitive mediation status is **CarrierNormal**. The imported bridge-witness closure certificate identifies the closed occupant of that status as $H_{\text{br/fix}}$, hence $C_H = H_{\text{br/fix}}$. $\square$

Corollary 8.5 (Carrier before representative). *A Higgs representative is standing-bearing only after carrier standing, route standing, or target update has been declared. No representative can precede C_H as a same-scope Higgs-domain primitive.*

9 Carrier minimality

Definition 9.1 (Carrier weakening). A carrier weakening is any same-scope submission that preserves the name C_H or “Higgs” while dropping at least one component of H_{role} , changing the invariant bundle I_D , or allowing a representative to select standing independently of carrier admission.

Theorem 9.2 (Carrier minimality). *No strict same-scope weakening of C_H preserves the fixed Higgs-domain target D_H . A submission that weakens C_H either changes target, loses Higgs-domain standing, imports hidden selector work, or collapses to governance-equivalent carrier notation.*

Proof. Consider a weakening. If it drops gauge inheritance, residual normal form, vacuum support, deformation mediation, mass-response standing, identity continuity, standing-preservation across the transition, or redescription stability, then by the role-bundle necessity lemma it no longer preserves

D_H . If it changes I_D , then it changes the admissibility boundary and exits the same fixed domain. If it allows a representative to select standing, then it imports a selector at the boundary, violating the AMetric/no independent standing generator/no transfer-carrier interface. If it changes none of these admissibility-relevant features, it contributes no distinct governance work and is governance-equivalent to C_H . These cases exhaust the possible same-scope weakenings. $\square$

Corollary 9.3 (No cost-free Higgs weakening). *One cannot keep the same Higgs-domain target while treating the Higgs as merely a scalar, merely a vev, merely a pole mass, merely a potential minimum, merely an RG feature, merely a vacuum route, or merely a thermal profile. Each weakening either discards a load-bearing role or silently reimports it.*

10 Representative collapse and primitive deletion

Definition 10.1 (Primitive representative claim). A primitive representative claim is a claim-form

$$R \Rightarrow \text{PrimitiveStand}_{D_H}(C)$$

where R is a Higgs representative and the claim attempts to use R as primitive Higgs-domain standing without carrier descent, route normal form, or target update.

Theorem 10.2 (Primitive representative collapse). *For every familiar Higgs representative R in Table 5,*

$$R \not\Rightarrow \text{PrimitiveStand}_{D_H}(C).$$

Equivalently, the primitive survivor status of representative-grounded Higgs standing is zero:

$$\text{Surv}_{D_H}^{\text{Primitive}}(C_R) = 0.$$

Proof. Let R be a representative. If R is only writable or calculationally useful, it has availability but not standing. If R is gauge-, scheme-, normalization-, matching-, redescription-, or input-set-dependent, then it fails quotient stability. If R is a measured response record, it is witness-bearing but not carrier-generating. If R is a downstream route object, such as a vacuum branch or thermal profile, it is evaluated by a route-specific quotient rather than by primitive bridge standing. If R becomes standing-bearing through C_H , route closure, or target update, then R is no longer primitive. Thus every primitive representative claim collapses. $\square$

Corollary 10.3 (No primitive Higgs scale). *A Higgs scale value or scale coordinate does not supply primitive Higgs-domain standing. Scale content is carrier-derived, route-derived, boundary-normalized, or target-updating.*

Proof. A scale may be a vev coordinate, pole-mass record, RG value, threshold parameter, hierarchy display, or metric-projection display. Each is a representative or downstream route object. None supplies H_{role} by itself. Therefore primitive scale standing is deleted in Paper 1. $\square$

Corollary 10.4 (No primitive Higgs vacuum). *A Higgs-vacuum branch does not supply primitive Higgs-domain standing. Vacuum standing belongs to a downstream endpoint route governed by static, lifetime, thermal, or target-update certificates.*

Proof. This is the Paper-1 carrier version of the lower-vacuum endpoint theorem. The vacuum branch may be diagnostic or route-bearing after quotient descent, but it does not define the bridge/fixation carrier. [12] $\square$

11 Local Higgs normal form

Definition 11.1 (Higgs status classifier). For a same-scope Higgs-domain claim protocol P , define the fail-closed status classifier

$$\rho_{\text{status}}^H(P) \in \{\text{CarrierNormal}, \text{RepresentativeOnly}, \text{RouteNormal}, \\ \text{RouteIncomplete}, \text{TargetUpdate}, \text{StandingSmuggling}\}.$$

The classifier is assigned by the following priority order:

$$\begin{aligned} &\text{TargetUpdate} \succ \text{StandingSmuggling} \succ \text{RepresentativeOnly} \\ &\quad \succ \text{RouteIncomplete} \succ \text{RouteNormal} \\ &\quad \succ \text{CarrierNormal}. \end{aligned}$$

Its clauses are:

TargetUpdate P changes carrier inventory, target, invariant bundle, admissibility boundary, or source state.

StandingSmuggling P treats representative availability, empirical success, route availability, or later precision as standing before closure.

RepresentativeOnly P supplies only a scalar, vev, potential, pole-mass, RG, Yukawa, mass-matrix, vacuum, or thermal representative.

RouteIncomplete P preserves D_H but lacks a route certificate or carrier closure component required for its claim.

RouteNormal P is a downstream route-normal claim preserving C_H but evaluated by scale, vacuum, thermal, or extension-specific route maps.

CarrierNormal P preserves D_H , supplies H_{role} , closes the bridge witness package, and factors through C_H .

A carrier-normal or route-normal protocol may contain scalar, vev, potential, mass, RG, vacuum, or thermal representatives. It is classified as **RepresentativeOnly** only when no carrier certificate or route certificate is supplied.

Definition 11.2 (Local Higgs normal form). The local Higgs normal form is

$$R_H = (D_H, C_H, H_{\text{role}}, \mathcal{R}_H, \sim_H, \text{SF}_H),$$

where D_H is the fixed target, C_H the bridge/fixation carrier, H_{role} the required role bundle, $\mathcal{R}_H$ the representative class, $\sim_H$ the same-scope representative/carrier quotient, and SF_H the fail-closed status classifier.

Definition 11.3 (Partial Higgs normalizer). The partial normalizer $\text{NF}_H(P)$ is defined, written $\text{NF}_H(P) \downarrow$, exactly when a claim protocol P preserves D_H , supplies the role bundle H_{role} , supplies the bridge witness package needed by C_H , and imports no representative selector or downstream route as primitive carrier standing. When defined,

$$\text{NF}_H(P) = R_H$$

with components instantiated by P 's declared data.

Theorem 11.4 (Higgs normal-form theorem). *Every same-scope Higgs-domain standing claim factors through the local Higgs normal form:*

$$\text{SameScopeHiggsStanding}_{D_H}(P) = 1 \Rightarrow \text{NF}_H(P) \downarrow$$

or P is a downstream RouteNormal claim over C_H .

Primitive carrier standing survives only in the carrier-normal case.

Proof. Let P be a same-scope Higgs-domain claim. The status classifier is fail-closed by priority. If P changes target or invariant bundle, it is **TargetUpdate**. If it converts representative availability or downstream success into standing, it is **StandingSmuggling**. If it supplies only a representative, it is **RepresentativeOnly**. If it preserves target but lacks a carrier or route component, it is **RouteIncomplete**. If it is a scale, vacuum, thermal, or extension-specific route over an already admitted carrier, it is **RouteNormal**. Only if none of those non-carrier statuses applies does P preserve D_H , supply H_{role} , close the bridge witness package, and factor through C_H ; then $\text{NF}_H(P) \downarrow$ and P is **CarrierNormal**. These cases exhaust the local governance-bearing loci by fixed-domain exhaustion. $\square$

Corollary 11.5 (Route availability is not carrier identity). *A route-normal claim over C_H , such as a hierarchy display, vacuum-lifetime verdict, thermal transition verdict, or extension descent, does not establish primitive Higgs carrier identity. It is downstream use of a carrier, not replacement of the carrier.*

12 Initial exhaustion register

Table 9: Initial Paper-1 Higgs-domain exhaustion register.

Claim-form	Status	Result
Primitive scalar Higgs	RepresentativeOnly or StandingSmuggling	Deleted as primitive; may present C_H after carrier closure.
Primitive vev Higgs	RepresentativeOnly	Deleted as primitive; vev-like coordinates are derivative or boundary-normalized.
Primitive potential Higgs	RepresentativeOnly	Deleted as primitive; potential charts do not select carrier standing.
Primitive pole-mass Higgs	RepresentativeOnly	Deleted as primitive; pole mass is response witness, not bridge carrier.
Primitive RG-scale Higgs	RepresentativeOnly	Deleted as primitive; RG labels are transport/diagnostic.
Primitive Yukawa-form Higgs	RouteIncomplete	Yukawa compatibility is downstream of carrier standing.
Primitive mass-matrix Higgs	RouteIncomplete	Mass matrices require downstream deformation-response ledgers.
Primitive vacuum-branch Higgs	RouteNormal or RepresentativeOnly	Vacuum claims are downstream endpoint routes, not carrier primitives.

Claim-form	Status	Result
Primitive thermal-transition Higgs	RouteNormal	Thermal claims are downstream finite-temperature route statuses.
Higgs bridge/fixation carrier	CarrierNormal	Paper-1 primitive survivor under the imported bridge-witness closure certificate.
BSM/composite/extra- scalar Higgs	TargetUpdate or CarrierNormal	Same carrier if role-preserving and descending to C_H ; target update otherwise.

Theorem 12.1 (Initial exhaustion theorem). *Inside D_H , the initial representative claim-forms in Table 9 are fully classified. All primitive representative claim-forms are deleted. The only Paper-1 primitive survivor is the bridge/fixation carrier C_H , relative to the imported bridge-witness closure certificate.*

Proof. Each row is classified by the Higgs status classifier. Representative-only rows have availability but no primitive standing. Route rows are downstream and therefore do not supply carrier identity. Target-update rows leave the same-scope target unless they descend back to the same carrier role. The carrier-normal row survives by the carrier admissibility theorem, the mediation alternative exhaustion lemma, the bridge-carrier necessity theorem, and the imported bridge-witness closure certificate. Since the rows exhaust the familiar same-scope primitive representative routes and fixed-domain exhaustion excludes a fifth governance locus, the register is complete for Paper 1. $\square$

13 Relation to standard Higgs and Standard Model language

The result is not anti-Higgs, anti-Standard-Model, or anti-field-theory. It does not deny that the Standard Model Higgs is successfully represented by scalar-field language, that electroweak symmetry breaking is physically meaningful, that the Higgs excitation has measured response records, that Yukawa structures are central to fermion mass-response, or that effective potentials and RG tools are essential for vacuum and hierarchy questions. It denies only the promotion of those representatives into primitive same-scope Higgs-domain standing.

Standard electroweak and Higgs formalism supplies the comparison language for this paper. The physics-facing details are inherited through the cited Standard Model, gauge-vacuum, renormalization, Higgs-bridge, hierarchy, and vacuum-route support papers. The present manuscript adds the carrier-status classification, not a replacement physics model.

Proposition 13.1 (Physics idiom translation). *The statement “the Higgs is a scalar field with a vev and an excitation” translates in this paper as:*

scalar/vev/excitation language is an admitted representative package for C_H in the Standard Model target.

It does not translate as:

scalarity or vev-status is primitive Higgs identity.

Proof. The first translation preserves representative use after carrier standing is fixed. The second reverses the dependency order and treats representative notation as carrier selector. The representative insufficiency theorem blocks that promotion. $\square$

Proposition 13.2 (No denial of empirical anchors). *Measured Higgs-sector quantities can be witness-bearing and physically indispensable while remaining non-primitive as identity grounds. Empirical anchors support routes and calibrations; they do not by themselves generate C_H .*

Proof. Empirical anchors enter as response records or boundary-normalized inputs. By the no independent standing generator/no transfer-carrier/no same-act repair principle, success or measurement cannot generate standing independent of the admissibility gate. If the anchor is used through an admitted carrier or route, its role is preserved; if it is used as primitive selector, the use is standing-smuggling. $\square$

14 Handoff to the later Higgs arc

14.1 Paper 2: Higgs structural identity

Paper 1 establishes carrier normal form. Paper 2 receives the identity question:

What makes something *the Higgs* rather than Higgs-associated representative content?

The Paper-2 theorem target is the uniqueness/exhaustion claim that every same-scope Higgs identity claim must preserve C_H , and any alternative occupant of the required role either reduces to C_H or declares target update.

14.2 Paper 3: Higgs scale and hierarchy

The scale/hierarchy paper treats metricity as downstream projection surface rather than as foundation. Paper 1 supplies the numerator carrier C_H ; later hierarchy work evaluates scale projections or carrier-order displays without substituting vev, pole mass, RG scale, or Planck label as primitive anchors. [11]

14.3 Paper 4: Higgs vacuum and stability

The vacuum paper classifies lower branches, metastability, lifetime routes, thermal crossover, first-order transition deletion, and minimal-SM EWBG route deletion as downstream endpoint questions. Paper 1 supplies the carrier-level reason why a vacuum branch cannot define Higgs identity. [12]

14.4 Paper 5: Extensions and extra scalars

The extension-classifier paper tests whether singlets, second doublets, composite sectors, dilaton-like carriers, supersymmetric thresholds, SMEFT terms, or hidden sectors preserve the same bridge/fixation role, add representatives over C_H , or change target. Paper 1 supplies the status classifier; Paper 2 supplies identity uniqueness.

14.5 Paper 6: Higgs endpoint synthesis

The synthesis paper states the final domain endpoint: Higgs-domain standing is centered on C_H ; scalar, vev, mass, scale, vacuum, thermal, metric, and extension descriptions are representative or route statuses; and same-scope identity is exhausted by unique bridge-carrier occupation.

15 Same-target closure and objection discipline

15.1 Denial obligations

A same-target objection to Paper 1 must do at least one of the following:

1. show that target-relative degeneracy misclassifies ordinary physics work rather than only same-scope standing claims;
2. show a same-scope non-degenerate construction that preserves target identity, stephood, act-time finality, and redescription fidelity while avoiding the kernel roles;
3. show that an internal metric, fit score, naturalness measure, or RG coordinate can license boundary standing without becoming a selector or target update;
4. show that downstream success, route availability, or later precision can generate primitive Higgs standing without violating standing conservation or same-act irreversibility;
5. break the kernel import or show that claim-bearing Higgs-domain use does not require reference, standing, admissibility, or irreversibility;
6. break the admissibility consequence imports: AMetric boundary, bivalence, standing identity, conservation, unique interior, or no independent standing generator/no transfer-carrier/no same-act repair;
7. show that D_H smuggles C_H into the target definition;
8. show that the local definitions of order-to-deformation transition or mass-response readiness import the carrier conclusion;
9. identify a component of H_{role} that is not required for same-scope Higgs-domain standing;
10. supply a representative that carries H_{role} without reducing to C_H , route normal form, or target update;
11. break the mediation alternative exhaustion lemma or its mapping to fixed-domain exhaustion loci;
12. break the imported bridge-witness closure certificate or one of its stated closure dependencies;
13. break the bridge-carrier necessity theorem;
14. break the carrier minimality theorem;
15. exhibit a genuine sixth mediation class or fifth same-scope governance-bearing locus;
16. provide a faithful same-target counterexample satisfying the conditions in Section 16.

15.2 Standing-objection register

Table 10: Standing-objection register for Paper 1.

Objection	Standing	Response
The Higgs is a scalar field.	Partial	Granted as representative language; denied as primitive identity/standing.
The vev defines electroweak breaking.	Partial	Granted as a useful coordinate/input; denied as carrier selector.
The pole mass is physical.	Granted	It is a witness-bearing response record, not the bridge/fixation carrier.
Yukawa couplings show the Higgs role.	Partial	They show downstream compatibility presupposing bridge standing.
The Higgs potential selects the vacuum.	Partial	Potential use is route/diagnostic; not primitive carrier standing.
The Standard Model already has a Higgs doublet.	Granted	Paper 1 classifies the standing role of the Higgs-domain carrier; it does not deny SM presentation.
An extra scalar may also be Higgs-like.	Compatible	It must either preserve/reduce to the same carrier role or declare target update.
Metric/gravity hierarchy is the deeper issue.	No for Paper 1	Metricity is downstream projection skin relative to the carrier problem.
Later precision may refine attributes.	Compatible	Precision can update routes or source state; it does not retroactively generate primitive representative standing.
A fifth primitive exists.	Serious if specified	It must preserve D_H , not reduce to C_H , route normal form, representative bookkeeping, or target update.

16 Faithful same-regime counterexample conditions

A faithful same-target counterexample to Paper 1 must preserve all of the following:

1. the fixed Higgs-domain target D_H ;
2. the kernel structure K_D ;
3. the admissibility consequence interface;
4. gauge inheritance and representation continuity;
5. residual/vacuum support;
6. deformation mediation and mass-response standing;
7. identity continuity and standing-preserving redescription;
8. the same-scope/richer-scope boundary;
9. the absence of hidden representative selectors;
10. fixed-domain exhaustion and the no-fifth-locus discipline.

It must then produce at least one of the following:

1. primitive Higgs-domain standing for scalarity, vev-status, potential-minimum status, pole-mass status, RG status, Yukawa-form status, mass-matrix status, vacuum-branch status, or thermal-transition status independent of C_H , route normal form, or target update;
2. a same-scope bridge alternative preserving H_{role} while not reducing to C_H ;
3. a fifth local governance-bearing locus not covered by boundary typing, standing partition, lawful quotient, or admissible continuation.

A construction that adds a new scalar, threshold, symmetry, UV rule, gravitational source, composite-ness carrier, empirical witness, or altered continuation clause is a target update unless it supplies a descent theorem back to C_H .

17 Conclusion

The first Higgs arc paper closes the carrier problem. A Higgs-domain claim cannot be standing-bearing merely because a scalar symbol is written, a vev coordinate is displayed, a potential minimum is chosen, a pole mass is measured, a running parameter is transported, a Yukawa form is available, a mass matrix is built, a vacuum branch is plotted, or a thermal profile is computed. Those objects are representatives, witnesses, diagnostics, or downstream routes. They become claim-bearing only by carrier descent, route normal form, or declared target update.

The standing-bearing survivor in Paper 1 is the Higgs bridge/fixation carrier

$$C_H := H_{\text{br/fix}}.$$

It is forced by the role-bundle requirement: gauge inheritance, residual/vacuum support, deformation mediation, mass-response standing, identity continuity, standing-preservation across the transition, and redescription stability. Weakening the carrier loses the target, imports hidden selector work, changes scope, or collapses to governance-equivalent notation.

The paper therefore supplies the correct foundation for the next arc step. Paper 2 begins from the carrier theorem proved here, not from the questions “is the Higgs a scalar?” or “which numerical attribute defines the Higgs?” The next endpoint is:

Higgs identity = unique admissible occupation of the bridge/fixation carrier role.

A Claim and nonclaim register

ID	Statement	Status
HBC-0A	Paper 1 distinguishes target-relative degeneracy from criticism of ordinary physics work.	Orientation claim
HBC-0B	AASC constraints apply only to claims of non-degenerate same-scope standing.	Scope principle

ID	Statement	Status
HBC-0C	Non-degenerate construction is explained as target identity, admissible stephood, act-time finality, and redescription fidelity.	Orientation definition
HBC-0D	AMetric boundary discipline separates internal measures from boundary authority.	Audit table
HBC-0E	Downstream success and route availability cannot generate primitive Higgs standing.	Orientation consequence
HBC-N0A	Paper 1 does not classify ordinary physics calculations as degenerate merely because they are calculations.	Nonclaim
HBC-N0B	Paper 1 does not prohibit internal metrics, probabilities, fits, naturalness diagnostics, or uncertainty envelopes.	Nonclaim
HBC-1	Claim-bearing Higgs-domain use imports K_D .	Kernel theorem
HBC-2	Higgs-domain admissibility inherits AMetric, bivalent, conservation, and no same-act repair consequences.	Import theorem
HBC-3	D_H fixes a neutral order-to-deformation transition target and does not smuggle C_H .	Target theorem
HBC-3A	Order-to-deformation transition is locally defined as standing-preserving passage from order to deformation response.	Definition
HBC-3B	Mass-response readiness is locally defined without downstream numerical mass import.	Definition
HBC-4	H_{role} is required for same-scope Higgs-domain standing.	Role theorem
HBC-5	Mediation alternatives exhaust into representative, route, standing-smuggling, target-update, or carrier-normal classes.	Exhaustion lemma
HBC-5A	Mediation status classes are mapped to fixed-domain exhaustion loci.	Audit table
HBC-6	Imported bridge-witness closure identifies $H_{\text{br/fix}}$ as the closed bridge occupant.	Import theorem
HBC-6A	Imported closure dependencies are decompressed into selector exclusion, witness package, Standard Model instantiation, residual normal form, gauge-vacuum compatibility, and mass-response readiness.	Audit table
HBC-7	$C_H = H_{\text{br/fix}}$ is admitted as carrier under the imported bridge-witness closure certificate.	Carrier theorem
HBC-8	Scalarity does not supply primitive Higgs standing.	Collapse theorem
HBC-9	Vev-like coordinates do not supply primitive Higgs standing.	Collapse theorem
HBC-10	Potential minima do not supply primitive Higgs standing.	Collapse theorem
HBC-11	Pole-mass records do not supply primitive Higgs standing.	Collapse theorem

ID	Statement	Status
HBC-12	RG/running labels do not supply primitive Higgs standing.	Collapse theorem
HBC-13	Yukawa and mass-matrix structures are downstream.	Boundary theorem
HBC-14	Vacuum and thermal statuses are downstream routes.	Handoff theorem
HBC-15	$C_H = H_{\text{br/fix}}$ is the Paper-1 carrier survivor relative to the imported bridge-witness closure certificate.	Main theorem
HBC-N1	Paper 1 does not derive the Higgs vev.	Nonclaim
HBC-N2	Paper 1 does not derive Higgs pole mass or self-coupling.	Nonclaim
HBC-N3	Paper 1 does not prove full Higgs identity uniqueness.	Nonclaim/Paper 2
HBC-N4	Paper 1 does not rule out BSM Higgs sectors.	Nonclaim

B Source-lock and dependency map

Source family	Imported role	Use in Paper 1
Kernel of admissibility	Four-role kernel, non-derivability, governance equivalence, no same-act repair	Imports reference, standing, admissibility, and irreversibility for claim-bearing Higgs-domain use.
Structure of Admissibility	AMetric boundary, bivalence, standing as reuse-stable admissibility, unique interior, conservation	Supports representative collapse, no primitive selector results, and standing-preservation discipline.
Fixed-Domain Exhaustion	No fifth same-scope governance locus; boundary, standing, quotient, and continuation loci	Maps mediation statuses to fixed-domain loci and blocks any sixth same-scope mediation class.
Closure by Exhaustion	Same-scope operators are bookkeeping-equivalent or inadmissible	Blocks representative totalization and carrier replacement by operator success.
AMetric Boundary	No selector, no parameter, no primitive carrier, no cross-boundary transport	Supplies boundary discipline for vev, metricity, scale, typicality, and naturalness.
Impossibility Suite	No independent standing generators, no standing-transfer carriers, no same-act repairs	Supplies the standing-smuggling classifier and denial of repair import.
Standard Model structure	Structural SM target, gauge, vacuum, bridge/fixation, mass-response slots	Instantiates the electroweak comparison arena and locates the bridge/fixation role. [10]
Gauge-vacuum compatibility	Vacuum-mediated response and gauge-compatible standing	Supports $\mathbf{GVac}_D$, residual/vacuum support, and compatibility for bridge/fixation mediation. [7]
MGN3 Higgs bridge	Bridge-witness closure, canonical bridge carrier, selector exclusion, quotient reduction, response signature, reduced mediation map, bridge role-rank	Supports the imported closure certificate identifying $H_{\text{br/fix}}$.
Renormalization as scale transport	RG as transport, not primitive selector	Blocks running-label promotion. [9]

Source family	Imported role	Use in Paper 1
Higgs metric envelope	Scale-standing false-anchor elimination and carrier comparison handoff	Supports Paper 3 handoff and keeps metricity downstream.
Lower Higgs vacuum	Vacuum endpoint route classification and lower-branch non-promotion	Supports Paper 4 handoff and blocks vacuum-branch identity.

C Failure-route table

Route	Attempted promotion	Audit result
Scalar notation	Scalar field becomes Higgs identity	Representative only; standing requires C_H .
Vev selector	Vev chooses bridge carrier	Reverses dependency order; blocked.
Potential selector	Potential minimum chooses Higgs standing	Chart/route content; no primitive standing.
Pole-mass selector	Measured excitation defines carrier	Witness-bearing response, not primitive carrier.
RG selector	Running value fixes Higgs standing	Transport diagnostic; not carrier selector.
Yukawa import	Yukawa success selects bridge	Downstream import; MGN4-type question.
Mass-matrix import	Desired mass matrix selects bridge	Downstream import; not Paper-1 standing.
Vacuum branch	Lower branch or metastability defines Higgs	Downstream vacuum endpoint route; not carrier identity.
Thermal profile	Crossover/first-order status defines Higgs	Downstream finite-temperature route.
Metric projection	Hierarchy comparison defines Higgs	Downstream scale display; metricity is not carrier foundation.
Extra scalar	Additional scalar counted as same Higgs without role proof	Target update unless it descends to C_H .
Composite route	New substructure counted as same Higgs without descent	Target update unless it preserves the carrier role.

D Notation and glossary

Symbol	Meaning
K_D	Fixed-domain kernel structure $(K, \simeq_{\text{gov}}, I_D)$.
K	Governance role set $\{\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}\}$.

Symbol	Meaning
D_H	Fixed Higgs-domain electroweak order-to-deformation and mass-response transition target.
$D_{H,+}$	Richer-scope Higgs target after carrier/source/boundary update.
C_H	Higgs bridge/fixation carrier $H_{\text{br/fix}}$.
H_{role}	Required role bundle: gauge inheritance, residual/vacuum support, deformation mediation, mass response, identity continuity, standing-preservation across the order-to-deformation transition, redescription stability.
$\mathcal{R}_H$	Class of Higgs representatives: scalar, vev, potential, pole mass, RG, Yukawa, mass matrix, vacuum, thermal, etc.
R_H	Local Higgs normal form $(D_H, C_H, H_{\text{role}}, \mathcal{R}_H, \sim_H, \text{SF}_H)$.
SF_H	Fail-closed Higgs status classifier.
$\text{NF}_H(P)$	Partial normalizer defined when a claim protocol closes the carrier-normal form.
TargetUpdate	Status for target-changing claims.
StandingSmuggling	Status for claims converting availability/success into standing without closure.
RepresentativeOnly	Status for merely representative Higgs descriptions.
RouteNormal	Status for downstream route-normal use over the carrier.
CarrierNormal	Status for closed same-scope bridge/fixation carrier standing.

E Paper-2 handoff theorem template

The following theorem is not proved in this manuscript. It is recorded only as the natural theorem target for Paper 2.

Theorem E.1 (Higgs structural identity theorem template). *In any same-scope electroweak regime preserving D_H , K_D , H_{role} , and the Higgs normal form R_H , every Higgs identity claim factors through the unique admissible occupation of C_H . Scalarity, vev-status, potential-minimum status, pole-mass status, running-scale status, vacuum-branch status, thermal-transition status, and extension-content status are representative or route descriptions only.*

Status. Deferred to Paper 2. Paper 1 supplies the kernel import, role bundle, carrier normal form, and initial exhaustion register needed for the identity theorem. $\square$

F Theorem dependency index

This appendix records the load-bearing theorem dependencies. It is intended to make the manuscript auditable as a Paper-1 carrier theorem rather than as a narrative essay.

Result	Direct dependencies	Function in paper
Kernel trigger	Kernel import, claim-bearing use	Forces K_D for Higgs-domain claims.
Admissibility consequence import	Structure of Admissibility	Supplies AMetric, bivalence, standing identity, conservation.
No fifth local governance locus	Fixed-Domain Exhaustion	Blocks hidden same-scope carrier route.
No standing laundering	Impossibility Suite	Blocks success, fit, route availability, and later precision as standing sources.
Target non-smuggling	Neutral definition of D_H	Ensures carrier is not built into target.
Role-bundle necessity	D_H , kernel roles, SM structural target	Forces H_{role} .
Representative insufficiency	H_{role} , AMetric/no-promotion	Deletes primitive standing for representatives.
Mediation alternative exhaustion	D_H , H_{role} , fixed-domain exhaustion, status classifier	Closes representative, route, standing-smuggling, target-update, and carrier-normal alternatives.
Bridge-witness closure import	MGN3 bridge support, Standard Model structural support	Identifies $H_{\text{br/fix}}$ as the closed bridge occupant.
Carrier admissibility	Bridge-witness closure import, H_{role} , carrier criterion	Positively admits C_H .
Bridge-carrier necessity	Role bundle, bridge locus, mediation alternative exhaustion, bridge-witness closure	Proves same-scope primitive standing factors through the bridge-carrier normal form.
Carrier minimality	H_{role} , I_D , no-selector, target-update discipline	Blocks strict same-scope weakening.
Primitive representative collapse	Representative insufficiency, no standing laundering	Deletes primitive representative claim-forms.
Higgs normal-form theorem	Status classifier, fixed-domain exhaustion	Defines R_H as Paper-1 normal form.
Initial exhaustion theorem	Normal form, register table, carrier admissibility, bridge-witness closure	Delivers Paper-1 endpoint.

G Anti-circularity and forbidden-retyping audit

Possible circularity or forbidden retyping	Audit response
D_H smuggles in C_H .	False. D_H fixes comparison ledgers and candidate descriptions only. Carrier selection occurs only after role-bundle necessity and representative insufficiency.
C_H is defined as the surviving object.	False. C_H is defined as the bridge/fixation carrier role. Its survival is proved by carrier admission and exhaustion of representative primitives.
Representatives are deleted by definition.	False. Representatives are positively admitted as descriptions, witnesses, diagnostics, or routes. Only primitive representative standing is deleted.
Vev is denied physicality.	False. Vev-like coordinates may be physical, boundary-normalized, or useful; they are denied only primitive carrier authority.
Pole mass is denied physicality.	False. Pole mass is witness-bearing response content, not primitive bridge/fixation identity.
Yukawa/mass matrices are removed from physics.	False. They are deferred downstream because they presuppose bridge standing.
Vacuum stability is excluded.	False. Vacuum stability and metastability are downstream endpoint routes over the carrier.
Metricity/gravity is rejected.	False. Metric comparison is downstream projection; it is not foundational for Paper 1 carrier standing.
AASC vocabulary is optional.	Terminology is optional; governance work is not. Any same-target objection must preserve or replace the kernel roles without circularity.
BSM is ruled out.	False. BSM/composite/extra-scalar proposals are classified as same-carrier realizations, representative additions, or target updates.

H Formal status machine

Definition H.1 (Paper-1 Higgs status machine). The Paper-1 status machine is the map

$$\mathcal{M}_H(P) = (\rho_{\text{status}}^H(P), \text{NF}_H(P), \text{Report}(P)),$$

where ρ_{status}^H is the fail-closed classifier, $\text{NF}_H(P)$ is the partial normalizer, and $\text{Report}(P)$ is the strongest admissible report: representative availability, route availability, carrier standing, target update, or failure.

Theorem H.2 (Status-machine soundness). *For every submitted same-scope Higgs-domain protocol P , the report generated by $\mathcal{M}_H(P)$ does not exceed the closed standing status of P .*

Proof. If P is target-updating, the machine reports target update and does not treat it as same-scope survival. If P smuggles standing, the machine reports failure rather than closure. If P is

representative-only, the machine reports availability or bookkeeping. If P is route-incomplete, the machine reports incompleteness. If P is route-normal, the machine reports route standing only under the corresponding route certificate and not carrier identity. If P is carrier-normal, $\text{NF}_H(P) \downarrow$ and the carrier report is licensed. The priority order prevents over-reporting in overlapping cases. $\square$

Corollary H.3 (No overclaim). *The strongest Paper-1 report is*

$\text{Report}_{\max} = \text{carrier-normal standing for } C_H \text{ plus deletion of primitive representative standing.}$

It is not a numerical Higgs prediction and not the full Paper-2 identity theorem.

I Construction audit checklist

A same-scope Higgs-domain claim is audited by the following checklist before downstream use.

1. **Target:** Does the claim preserve D_H , or does it declare $D_{H,+}$?
2. **Kernel:** Are reference, standing, admissibility, and irreversibility preserved?
3. **Boundary:** Does the claim introduce any selector, metric, probability, typicality, canonical representative, or naturalness score as boundary authority?
4. **Carrier:** Does the claim factor through C_H , or does it try to replace C_H with a representative?
5. **Role bundle:** Are all components of H_{role} preserved?
6. **Representative:** If a scalar, vev, potential, pole mass, RG value, Yukawa term, mass matrix, vacuum branch, or thermal profile appears, is it typed as representative, route, witness, or target update?
7. **Route:** If the claim is downstream, what route certificate makes it route-normal?
8. **Scope:** Does any new carrier, field, threshold, UV rule, composite structure, cosmological condition, or gravitational rule change the target?
9. **Report:** Does the report stay within carrier standing, route status, representative availability, target update, or failure?
10. **Handoff:** Which later paper is allowed to use the result, and in what status?

J Formalization boundary

No paper-specific Lean or proof-assistant artifact is asserted as part of this deliverable. A formalization of this theorem targets the status architecture rather than Standard Model phenomenology. The formal objects are:

1. the imported kernel structure K_D ;
2. the fixed target D_H and richer-scope transition $D_H \rightarrow D_{H,+}$;
3. the role bundle H_{role} ;
4. the representative class $\mathcal{R}_H$;
5. the quotient $\sim_H$;

6. the fail-closed status classifier SF_H ;
7. the partial normalizer NF_H ;
8. the initial exhaustion register.

It does not attempt to mechanize the Standard Model, electroweak data, RG equations, lattice thermodynamics, or Higgs phenomenology in this paper.

K Extension protocol

Any later manuscript using Paper 1 declares:

1. whether it uses C_H as carrier, representative, route input, or target-update comparator;
2. whether the claim is same-scope or richer-scope;
3. which components of H_{role} are preserved;
4. which representatives are used and whether they are typed as skin, witness, tensor, route, or anchor;
5. whether the claim attempts to generate standing by empirical success, naturalness, typicality, or later repair;
6. the exact endpoint status: representative availability, route availability, carrier normality, identity endpoint, target update, or failure.

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